



BACHELOR OF COMPUTER SCIENCE (SOFTWARE ENGINEERING)

COURSE NAME AND CODE:

CSE3023 – WEB-BASED APPLICATION DEVELOPMENT (K1)

LAB 2

SEMESTER II SESSION 2025/2026

PREPARED BY:

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LAB MODERATOR:

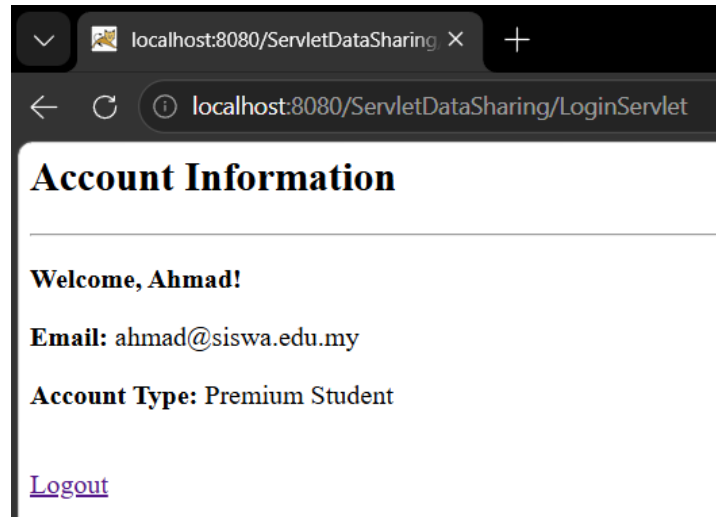
PN. WAN FATIN FATIHAH BINTI YAHYA

LAB 2

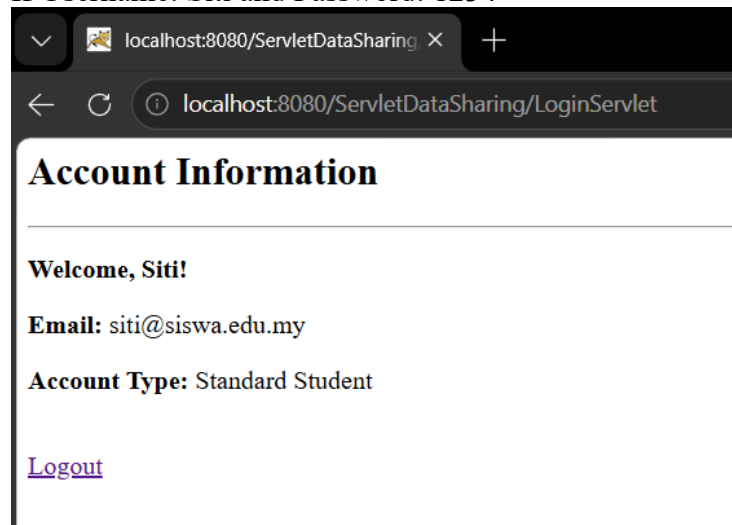
2.4 - Task 1: Data Sharing in Servlet

Output:

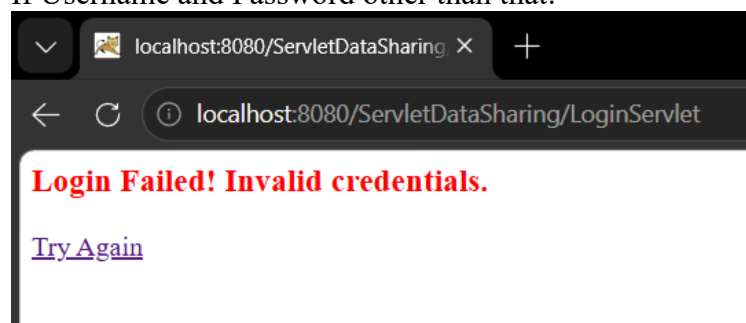
If Username: Ahmad and Password: 4567



If Username: Siti and Password: 1234



If Username and Password other than that:



Reflections

1. What have you learned from this exercise?

By completing this exercise, I have learned how to develop a dynamic web application that facilitates **data sharing** between multiple components by using the `request.setAttribute()` and `request.getAttribute()` methods. I also gained practical experience in using the `RequestDispatcher` for **request forwarding**, which allows one servlet to process authentication logic and then pass control to another servlet to display the results. Additionally, this task provided a foundational understanding of the **Model-View-Controller (MVC) workflow**, from capturing user input through an HTML form to processing that data via a back-end Java Servlet on an Apache Tomcat server.

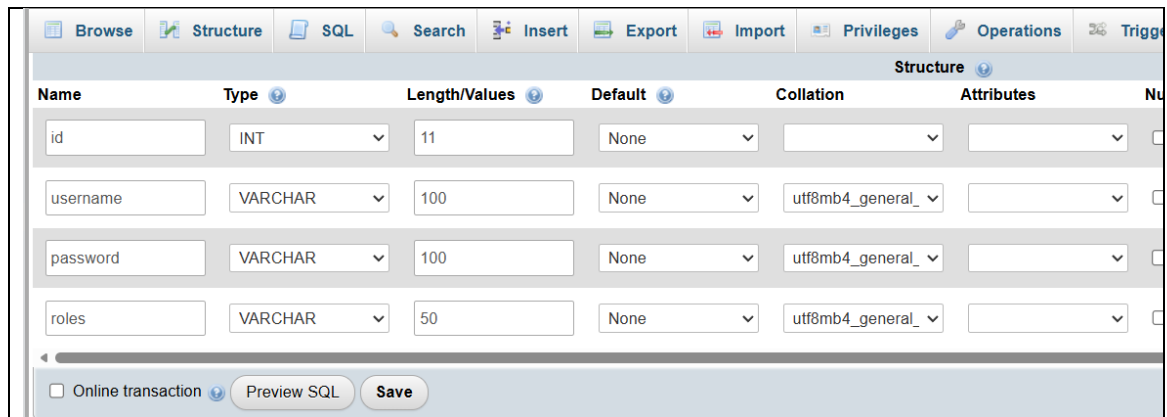
2. What are the common methods used in Java Servlet?

Based on the implementation steps and code provided in the module, the common methods used in Java Servlets include:

- **doPost(HttpServletRequest, HttpServletResponse):** Used to process data sent from an HTML form using the POST method, such as user login credentials or database entry data.
- **doGet(HttpServletRequest, HttpServletResponse):** Used to handle GET requests, typically for retrieving data or viewing records, such as displaying a user list or fetching a specific record by its ID.
- **init():** Overridden to initialize resources when a servlet is first loaded, such as instantiating a **DAO (Data Access Object)**.
- **getParameter(String):** Used to retrieve user input values from HTML form fields or URL parameters.
- **setAttribute(String, Object) and getAttribute(String):** Used for sharing data objects between different servlets during a single request cycle.
- **setContentType(String) and getWriter():** Used to set the response type (e.g., "text/html") and obtain a `PrintWriter` to send HTML content back to the browser.

2.5 – Task 2: Creating A Table in MySQL Database

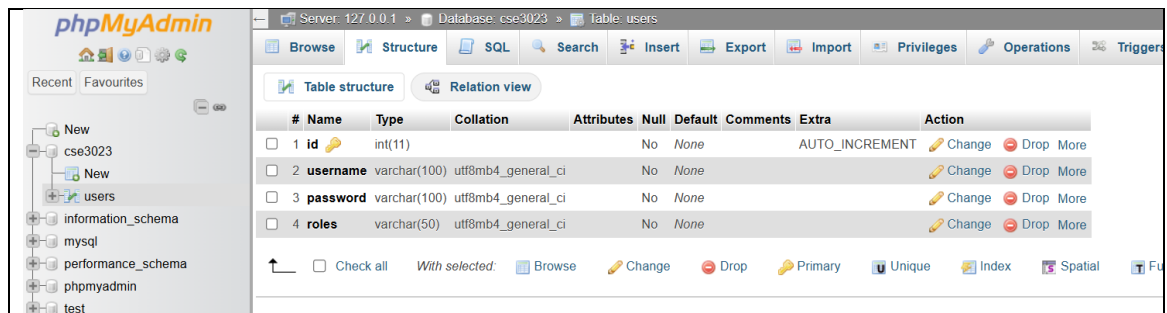
Input:



The screenshot shows the MySQL Structure window for creating a new table. The table has four columns: 'id' (INT, 11, None, utf8mb4_general_ci), 'username' (VARCHAR, 100, None, utf8mb4_general_ci), 'password' (VARCHAR, 100, None, utf8mb4_general_ci), and 'roles' (VARCHAR, 50, None, utf8mb4_general_ci). The 'Online transaction' checkbox is unchecked. The 'Preview SQL' and 'Save' buttons are visible at the bottom.

Name	Type	Length/Values	Default	Collation	Attributes	Null
id	INT	11	None	utf8mb4_general_ci		
username	VARCHAR	100	None	utf8mb4_general_ci		
password	VARCHAR	100	None	utf8mb4_general_ci		
roles	VARCHAR	50	None	utf8mb4_general_ci		

Output:

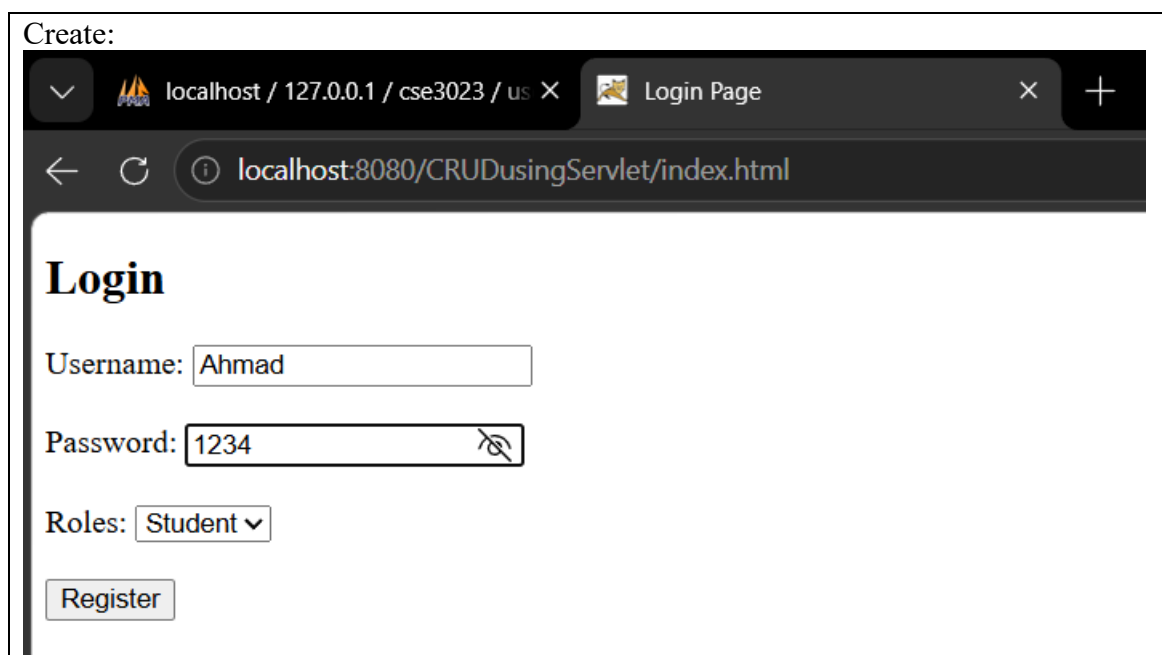


The screenshot shows the phpMyAdmin interface with the 'Table structure' tab selected for the 'users' table. The table structure is as follows:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
2	username	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
3	password	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
4	roles	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More

2.7 – Task 4: Using Servlets for Database CRUD Operation

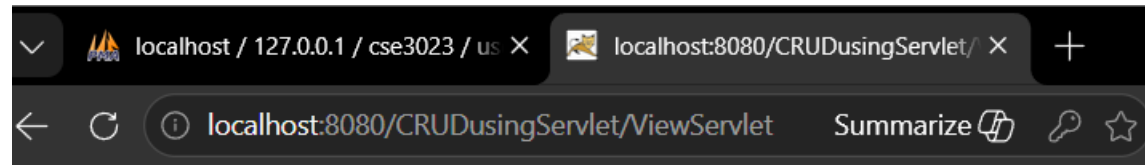
Output:



The screenshot shows a web browser window with the URL 'localhost:8080/CRUDusingServlet/index.html'. The page is titled 'Login' and contains a form with the following fields:

- Username: Ahmad
- Password: 1234 (with a toggle icon for visibility)
- Roles: Student (dropdown menu)
- Register button

Retrieve:

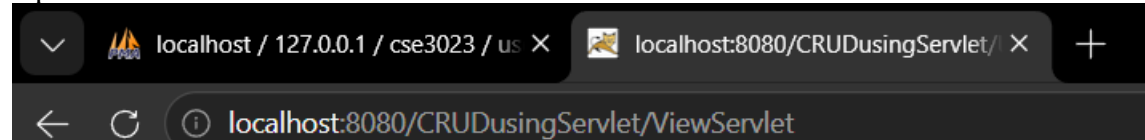


User List

ID	Username	Password	Role	Actions
1	Ahmad	1234	Student	Edit Delete

[Add New User](#)

Update:

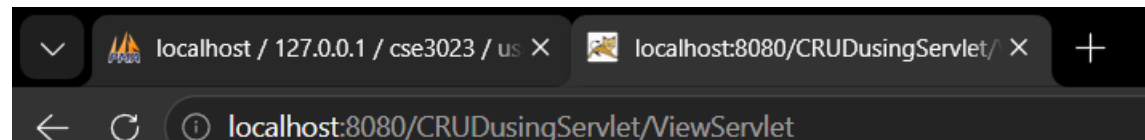


User List

ID	Username	Password	Role	Actions
1	Ahmadi	1234	Student	Edit Delete

[Add New User](#)

Delete:



User List

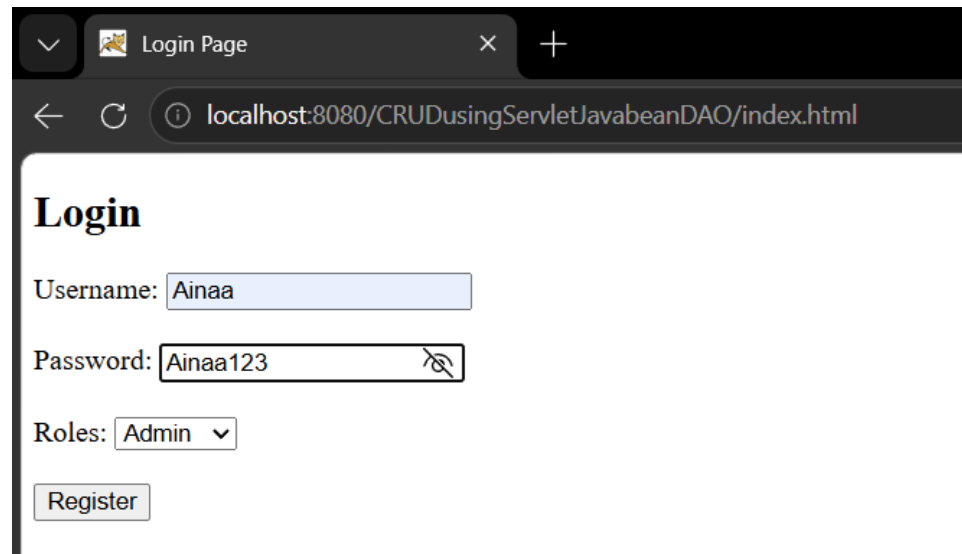
ID	Username	Password	Role	Actions
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[Add New User](#)

2.8 – Task 5: Using Servlets, DAO and Java Beans for Database CRUD Operations

Output:

Create:

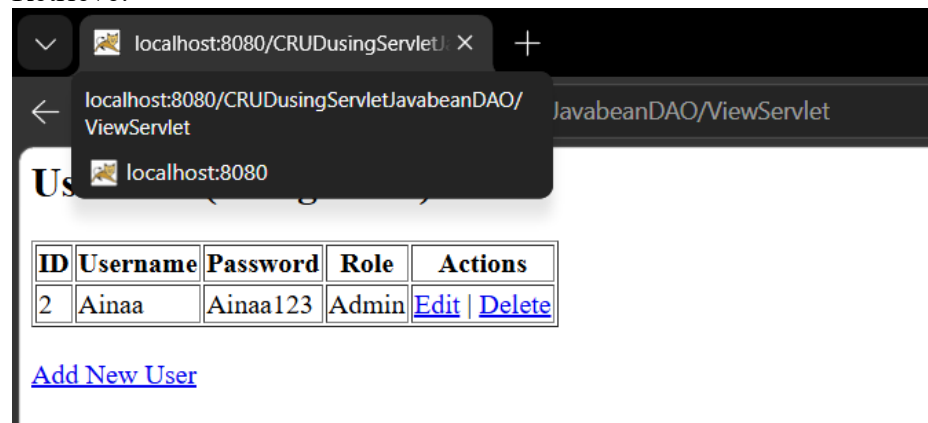


Username:

Password:

Roles:

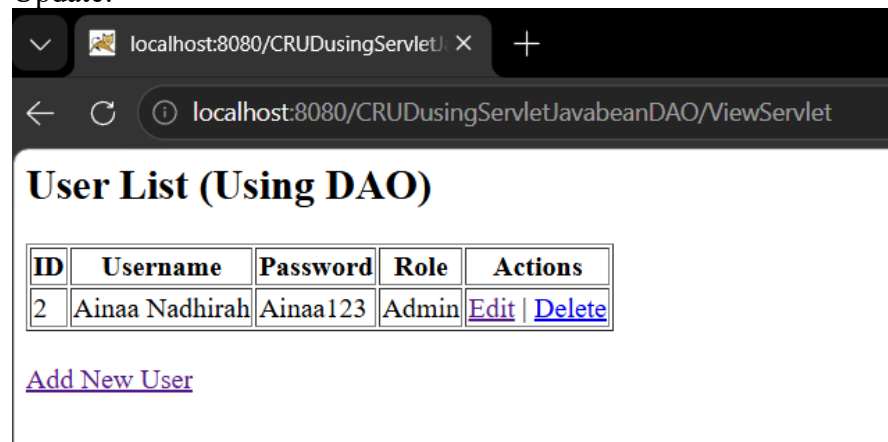
Retrieve:



ID	Username	Password	Role	Actions
2	Ainaa	Ainaa123	Admin	Edit Delete

[Add New User](#)

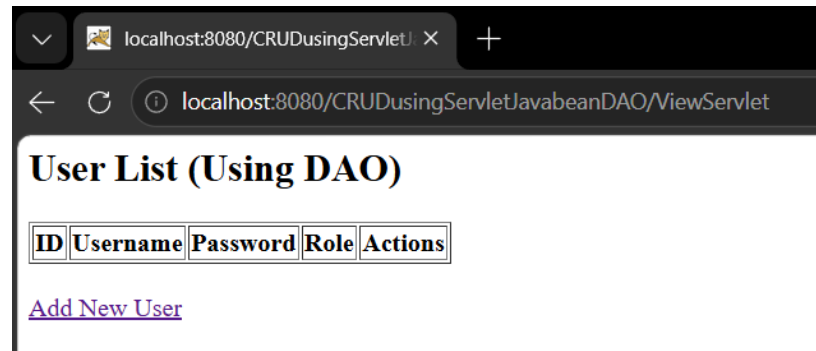
Update:



ID	Username	Password	Role	Actions
2	Ainaa Nadhirah	Ainaa123	Admin	Edit Delete

[Add New User](#)

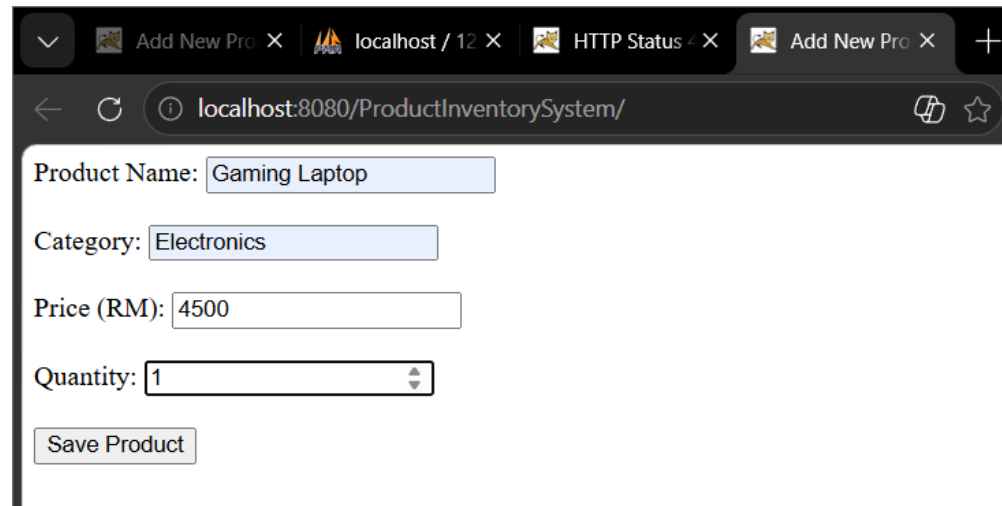
Delete:



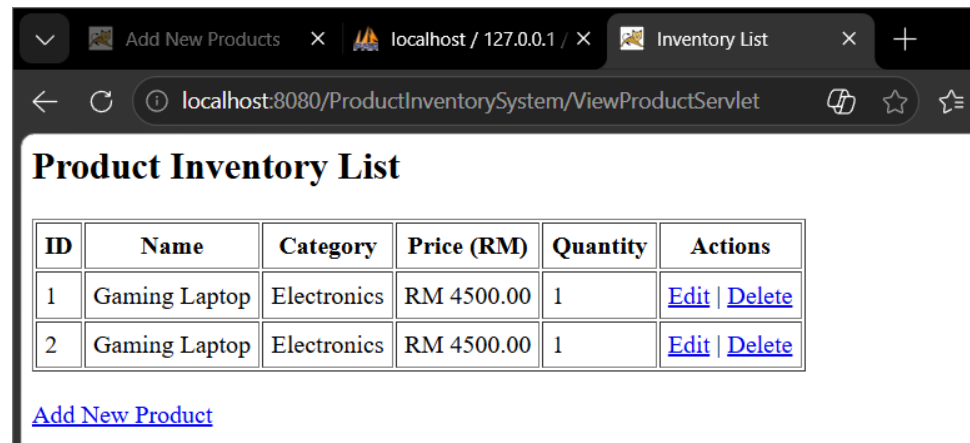
2.9 - Exercise:

Output:

Create:



Retrieve:



Update:

ID	Name	Category	Price (RM)	Quantity	Actions
1	Laptop	Electronics	RM 4500.00	1	Edit Delete
2	Gaming Laptop	Electronics	RM 4500.00	1	Edit Delete

[Add New Product](#)

Delete:

ID	Name
1	Laptop
2	Gaming Laptop

[Add New Product](#)

After click "OK":

ID	Name	Category	Price (RM)	Quantity	Actions
1	Laptop	Electronics	RM 4500.00	1	Edit Delete

[Add New Product](#)